

Monotone Phase Diagram in (τ , γ): high range

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Setup

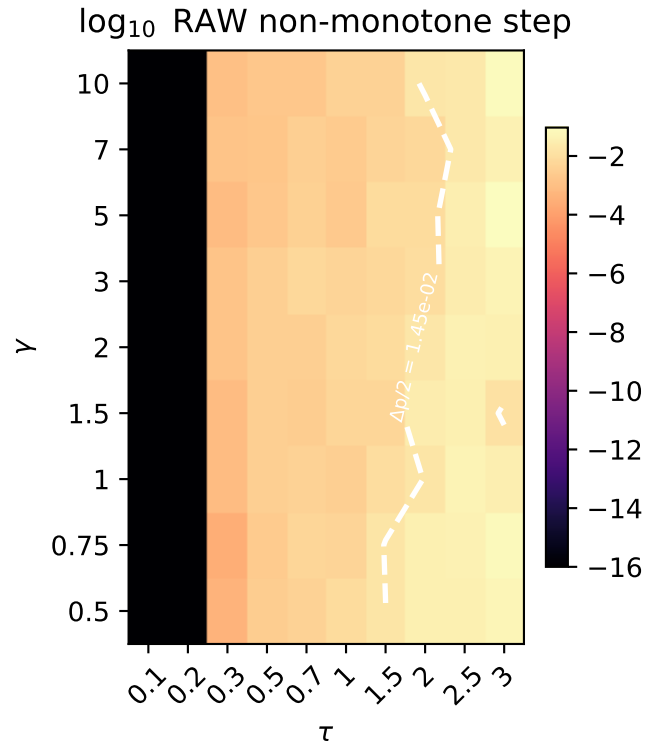
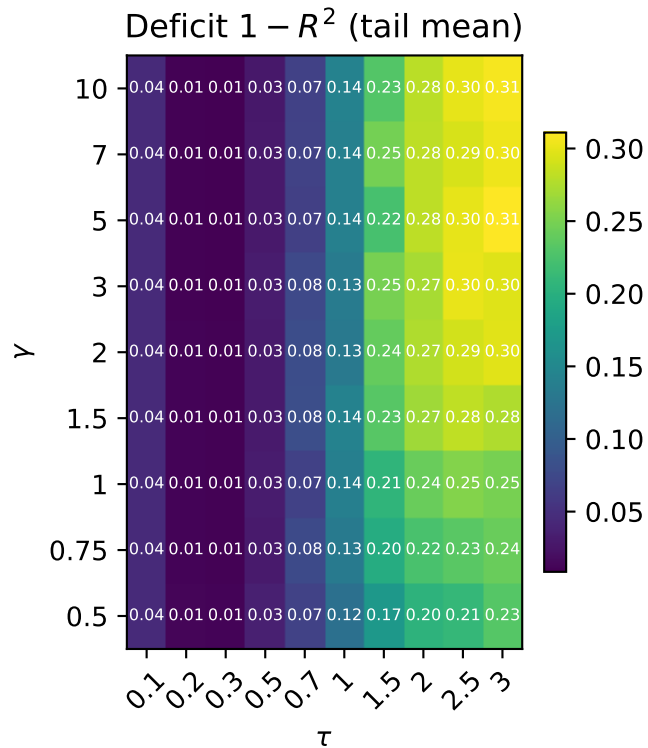
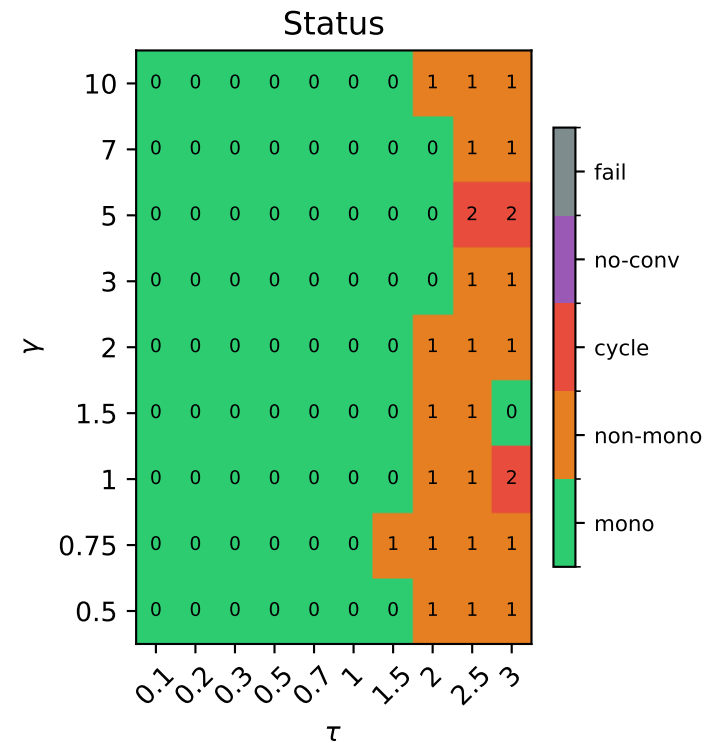
- * REZN economy: $K=3$ informed traders, CRRA-like utilities, Gaussian signals. No noise traders.
- * Endogenous-price (CMM) discrete-bin solver, $G = 21$, $M = 32$.
- * Sweep grid:
 - γ in $[0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 5.0, 7.0, 10.0]$
 - τ in $[0.1, 0.2, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 2.5, 3.0]$
- * For each (τ , γ) we run the soft-bin K-means iteration with monotone PAV projection ($\text{max_iter}=40$, $\text{h_p_factor}=1.5$) and record the final status:
 - 0: monotone_converged -- equilibrium is strictly monotone
(RAW non-monotonicity below $\Delta p / 2$)
 - 1: fixed_but_nonmonotone-- a fixed point exists but the RAW field has genuine non-monotonicity
 - 2: cycle -- iterate hits a 2-cycle
 - 3: not_converged -- ran out of iterations
 - 4: solver_failed -- numerical exception

Headline

Extends the (τ , γ) phase diagram into the upper-right corner. Confirms the original finding: the 'good zone' (strictly monotone equilibria) grows monotonically with risk aversion γ and shrinks monotonically with information variance τ .

- * 66 of 90 cells: monotone-converged.
- * 21 cells: fixed but non-monotone.
- * 3 cells: cycle.
- * 0 cells: not converged.
- * 0 cells: solver failed.

(tau, gamma) phase diagram — high range — G=21, M=32



Phase tables

Status matrix (0=mono, 1=non-mono, 2=cycle, 3=no-conv, 4=fail)

gamma \ tau		0.1	0.2	0.3	0.5	0.7	1	1.5	2	2.5	3

gamma= 0.5		0	0	0	0	0	0	0	1	1	1
gamma=0.75		0	0	0	0	0	0	1	1	1	1
gamma= 1		0	0	0	0	0	0	0	1	1	2
gamma= 1.5		0	0	0	0	0	0	0	1	1	0
gamma= 2		0	0	0	0	0	0	0	1	1	1
gamma= 3		0	0	0	0	0	0	0	0	1	1
gamma= 5		0	0	0	0	0	0	0	0	2	2
gamma= 7		0	0	0	0	0	0	0	0	1	1
gamma= 10		0	0	0	0	0	0	0	1	1	1

Deficit (tail mean of $1 - R^2$)

gamma \ tau		0.1	0.2	0.3	0.5	0.7	1	1.5	2	2.5	3

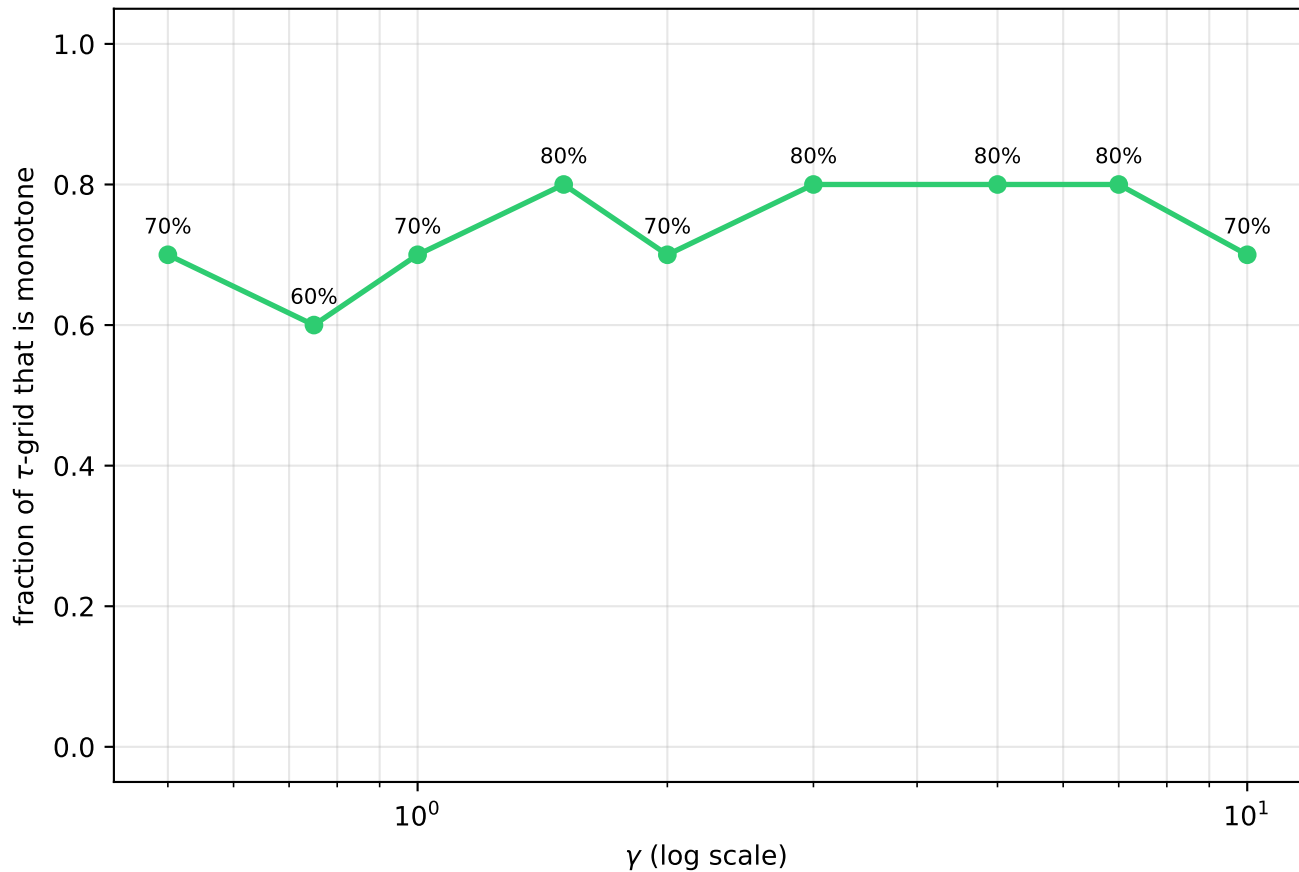
gamma= 0.5		0.04	0.01	0.01	0.03	0.07	0.12	0.17	0.20	0.21	0.23
gamma=0.75		0.04	0.01	0.01	0.03	0.08	0.13	0.20	0.22	0.23	0.24
gamma= 1		0.04	0.01	0.01	0.03	0.07	0.14	0.21	0.24	0.25	0.25
gamma= 1.5		0.04	0.01	0.01	0.03	0.08	0.14	0.23	0.27	0.28	0.28
gamma= 2		0.04	0.01	0.01	0.03	0.08	0.13	0.24	0.27	0.29	0.30
gamma= 3		0.04	0.01	0.01	0.03	0.08	0.13	0.25	0.27	0.30	0.30
gamma= 5		0.04	0.01	0.01	0.03	0.07	0.14	0.22	0.28	0.30	0.31
gamma= 7		0.04	0.01	0.01	0.03	0.07	0.14	0.25	0.28	0.29	0.30
gamma= 10		0.04	0.01	0.01	0.03	0.07	0.14	0.23	0.28	0.30	0.31

Wall time (seconds)

gamma \ tau		0.1	0.2	0.3	0.5	0.7	1	1.5	2	2.5	3

gamma= 0.5		8	6	4	11	11	12	16	20	31	24
gamma=0.75		7	6	4	9	12	15	17	27	317	31
gamma= 1		7	6	4	9	11	15	20	34	240	271
gamma= 1.5		6	6	4	9	13	14	19	23	37	108
gamma= 2		6	6	4	8	11	12	18	27	32	110
gamma= 3		6	6	4	8	10	12	21	28	50	243
gamma= 5		6	6	4	8	9	12	24	74	342	254
gamma= 7		6	6	4	8	9	12	35	39	310	253

Monotone share as a function of γ



Notes

Reading the figures

- * Status (panel 1): a green cell certifies a strictly monotone discrete-price equilibrium of the soft-K-means CMM solver. Orange cells reach a fixed point but the underlying RAW price-on-state field has genuine non-monotonicity above the $\Delta p / 2$ grid floor.
- * Deficit (panel 2): tail-mean of $(1 - R^2)$ of the logit price regression on $(u_1 + u_2 + u_3)$. Larger = farther from full revelation. Note that even strictly-monotone equilibria can carry sizeable revelation deficit at intermediate τ .
- * RAW violation (panel 3): max single-step non-monotonicity of the un-projected price field after iteration; the dashed $\Delta p / 2$ contour separates strict (green) from quasi (orange) monotonicity.

Limitations

- * Single-grid run ($G=21$, $M=32$); a publishable companion at higher resolution is the next step.
- * $\tau \geq 2$ with weak risk aversion is delicate -- the K-means iteration occasionally cycles instead of converging.